

Cage Structures

Ultra-High Metal-Ion Selectivity Induced by Intramolecular Cation- π Interactions for the One-Pot Synthesis of Precise Heterometallic Architectures

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Abstract: Heterometallic supramolecules, known for their unique synergistic effects, have shown broad applications in photochemistry, host-guest chemistry, and catalysis. However, there are great challenges to precisely construct heterometallic supramolecules rather than a statistical mixture, due to the limited metal-ion selectivity of coordination units. In particular, heterometallic architectures precisely encoded with different metal ions usually fail to form in a one-pot method when only one type of coordinated motif exists due to its poor metal-ion selectivity. Herein, we propose an effective intramolecular cation- π (IC π) strategy and successfully constructed the heterometallic supramolecule **Zn₂Cu₄L₃₄** by the one-pot self-assembly of tritopic terpyridyl ligand **L3** with Zn(II) and Cu(II), following a clear self-assembly mechanism in which only thermodynamic dimers **ZnL₁₂** and **Cu₂L₂₂** were constructed with model ligands **L1**, **L2**, Zn(II) and Cu(II) with perfect self-sorting and an ultra-high metal-selectivity feature. The successful construction of the heterometallic supramolecule **Zn₂Cu₄L₃₄**, in which the definite sequence of metal ions Zn(II) and Cu(II) is encoded in the one-pot method, will offer a novel approach to precisely construct heterometallic architectures.

Introduction

Among various non-covalent interactions, metal-ligand coordination interactions with highly directional and predictable features have been emerged to be one of the most powerful approaches for constructing highly intricate supramolecular architectures.^[1-21] As a favorable tool for mimicking complex biological systems and nature, multi-component coordination-driven self-assembly,^[22-39] which has evolved into an effective and order-out-of-chaos protocol for further achieving a single product with diversity, complexity, and functionality, has recently garnered increasing attention. Compared with others, the heterometallic supramolecules have unique synergistic effects^[40] and thus shown especially promising applications in medicine,^[41-42] photochemistry,^[43-44] host-guest chemistry,^[45-48] catalysis^[32, 49-53] and so on. It is crucial to precisely construct hetero-

metallic supramolecules^[54] in a highly controllable manner, rather than statistical mixture, which will be in favor of the development of functions and the revelation of structure-function mechanism.

Extensive efforts have been made in order to achieve precise construction of heterometallic supramolecules.^[48, 55-56] There are a series of strategies that can realize high efficiency and quantitative yield with one-pot synthesis procedures.^[55-56] However, these so-called “one-pot strategies” typically require the ligands to possess two or more coordination motifs with perfect self-sorting nature for different metal ions to give well-defined structures.^[57-60] Otherwise, the ligands with poor metal-selectivity will give chaotic products with disorderly metal arrangements (Path I in Scheme 1a). Consequently, it must rely on the time-, cost-, and labor-consuming stepwise self-assembly methods to construct heterometallic architectures,^[12, 51, 53, 62-75] in which

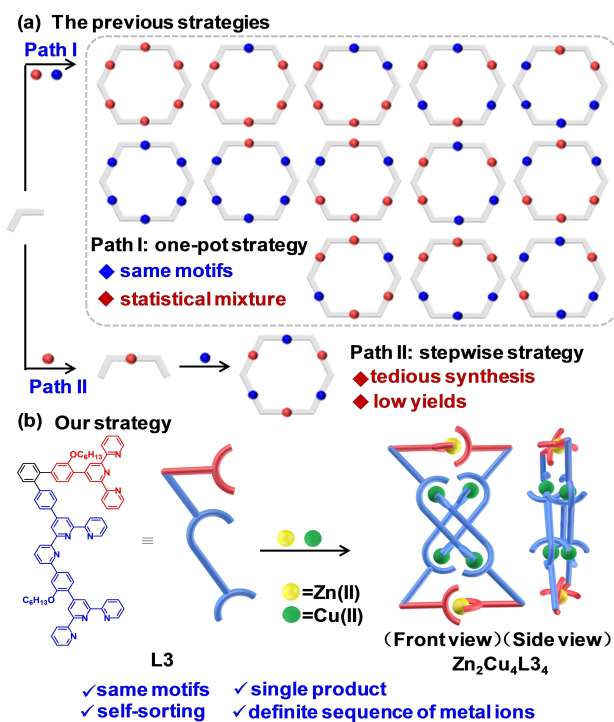
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Scheme 1. (a) Self-assembly of heterometallic supramolecules only with one type ligand via common one-pot strategy; (b) Self-assembly of the heterometallic supramolecule $\text{Zn}_2\text{Cu}_4\text{L}_{34}$ with ligand **L3**, Zn(II) and Cu(II).

metalloligands with additional unoccupied pendant binding sites can coordinate to the second set of more labile metal ions to give heterometallic architectures without mismatched coordination motifs (Path II in Scheme 1a). To date, it is still in their infancy and extremely challenging to furnish single heterometallic architectures with precisely ordered arrangements of metal ions using only one perfect metal-selective motif in one-pot strategy.^[56]

In this work, we developed such a novel but effective intramolecular cation- π strategy (named IC π) for precisely constructing heterometallic architectures in one pot. Based on this strategy, the heterobimetallic supramolecule $\text{Zn}_2\text{Cu}_4\text{L}_{34}$ was successfully prepared by one-pot self-assembly of a tritopic terpyridyl ligand **L3** with Zn(II) and Cu(II), which realized the definite sequence of metal ions in a highly error-free molecular programming manner (Scheme 1b). **L3** was combined from two model ligands of **L1** and **L2**. Notably, **L2** contained two terpyridyl motifs that could provide extra intramolecular cation- π interaction after complexation. The self-assembly experiments involving the model ligands **L1** and **L2** with Zn(II) and Cu(II) demonstrated that only the thermodynamic dimers ZnL_2 and Cu_2L_2 were formed, without any byproducts, which proved the mechanism of excellent self-sorting and metal-selectivity feature. Overall, the IC π strategy shows a clear competitive advantage by providing a straightforward and efficient method to construct heterobimetallic assemblies through a one-pot synthesis. This innovative approach opens a new avenue for the design and synthesis of complex

supramolecular architectures with precise metal-ion arrangements.

Results and Discussion

The self-assembly of mixed model ligands **L1 and **L2** with one type of metal ion.** Ligand **L3** with one common terpyridyl motif and one ditopic disymmetrical terpyridyl motif was strategically designed (Scheme 1). A phenyl group was introduced into the center of ditopic terpyridyl motif, which may produce new intramolecular interactions after coordinated with metal ions, change coordination micro-environment and further achieve metal-selectivity. To better understand self-assembly process of ligand **L3**, ligand **L3** was split into model ligands **L1** and **L2** (Figure 1). Ligand **L2** with two terpyridyl units was designed to self-assemble in head-to-tail form. Initially, the assemblies of ligand **L1** or **L2** with metal ions were obtained to serve as references and characterized by multiple nuclear magnetic resonance spectra (NMR, ^1H and ^{13}C) and Electrospray Ionization Mass Spectrometry (ESI-MS, Figure S43–S44). Further, the common tpy ligand **L1** and ditopic ligand **L2** were mixed in a 1:1 molar ratio to study the self-sorting behavior by adding Zn(II) with different ratios (Figure 1). The ^1H NMR revealed that, with metal ratio up to 2, only one set of signals corresponding to Zn_2L_2 was observed (Figure 1b),

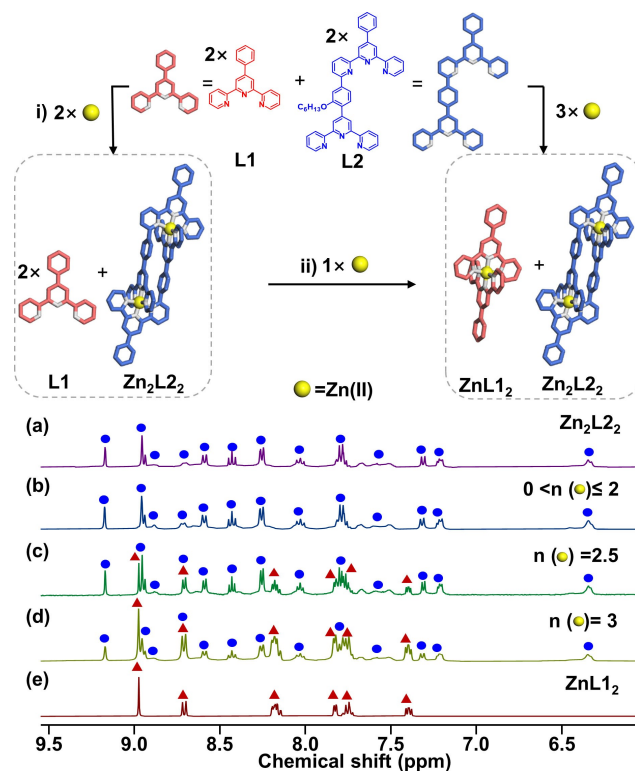


Figure 1. Competitive coordination experiments of ligands **L1**, **L2** with Zn(II) metal ions: ^1H NMR spectra (400 MHz, 298 K) of (a) Zn_2L_2 , (b–d) mixture of **L1**, **L2**, and different equivalents of Zn(II) metal ions, and (e) ZnL_2 in CD_3CN .

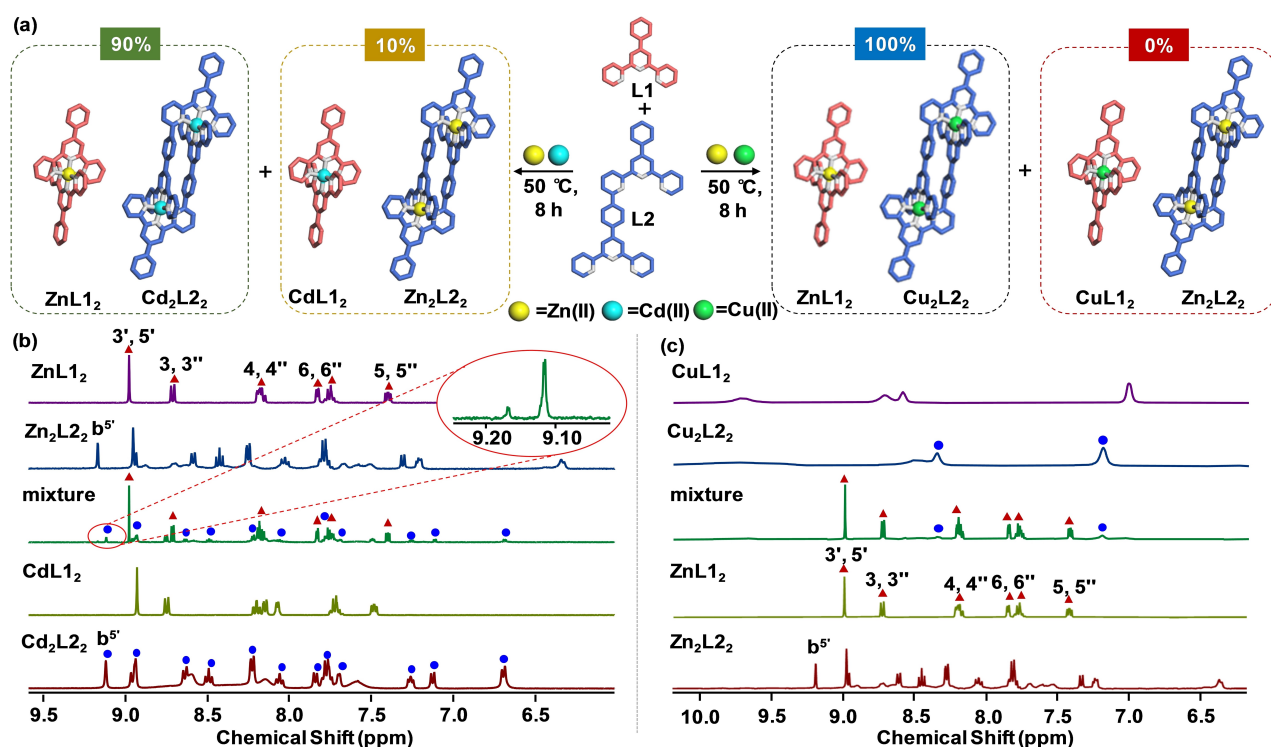


Figure 2. Selective coordination experiments of mixed ligands **L1**, **L2** with (a) Zn(II), Cd(II) and Zn(II), Cu(II). ^1H NMR spectra (500 MHz, 298 K) of the mixture of (b) **L1**, **L2**, Zn(II) and Cd(II), (c) **L1**, **L2**, Zn(II) and Cu(II).

indicating that Zn(II) preferentially coordinates with **L2**. As the metal ratio increased, a second set of peaks corresponding to **ZnL1₂** appeared (Figure 1c and 1d). This result confirms that the coordination capacity of **L2** takes priority over that of **L1**, and no mismatched products were observed during self-assembly process, which should be attributed to the presence of extra intramolecular forces. The same experimental results were obtained when Zn(II) was replaced with either Cd(II) or Cu(II), further validating the hypothesis and the general applicability of this approach (as shown in Figures S1 and S2). This consistent outcome across different metal ions reinforces the role of the intramolecular forces in driving the selective self-assembly process, demonstrating the robustness of this strategy for creating well-defined heterometallic supramolecular structures.

The self-assembly of model ligands **L1 and **L2** with two types of metal ions.** After clarifying the self-assembly mechanism with one type of metal ion, the self-assembly of mixed ligands **L1** and **L2** in the presence of different metal ions is investigated. Firstly, Zn(II) and Cd(II) were selected to coordinate with **L1** and **L2** in an exact stoichiometric ratio (**L1**:**L2**:Zn:Cd=2:1:1:1) in one pot and detected with NMR spectrum and ESI-MS (Figures 2a, 2b, S3, and S4), there is a mixture of complexes **ZnL1₂**, **CdL2₂**, **CdL1₂**, and **ZnL2₂**. From NMR spectrum, the integral area proportion of tpy- H^{b5} for **ZnL2₂** and **CdL2₂** is 1:9 (Figure S3), thus the proportion of **ZnL1₂**+**CdL2₂** and **CdL1₂**+**ZnL2₂** were 90% and 10%. And the self-assembly of mixed ligands **L1** and **L2** with Cd(II) and Cu(II) is also a mixture of **CuL1₂**, **CuL2₂**, **CdL1₂** and **CdL2₂** (Figure S16). The peaks corre-

sponding to tpy- $\text{H}^{3,5'}$ showed well-resolved and characteristic signals (Figure 2c), and the two sets of peaks belong to **CuL2₂** and **ZnL1₂**, indicating a precise and complete self-sorting behavior for Cu(II) and Zn(II) (Figure S5).

To further elucidate the mechanism of metal-ion selectivity, the binding constants (*K*) of ligands **L1** and **L2** with metal ions Zn(II), Cd(II), and Cu(II) were measured by the UV/Vis titration (Figures S7–S13) and summarized in Table 1. For the same type of metal ion, **L2** exhibits higher binding constant than **L1**, which is consistent with the experimental phenomenon (Figure 1). The binding constants of the three metal ions with **L1** were found to follow the descending order: $K_{\text{Zn}(\text{L1})_2} > K_{\text{Cu}(\text{L1})_2} > K_{\text{Cd}(\text{L1})_2}$, which is consistent with the reported literature.^[76] However, the ordering of the binding constants of **L2** was $K_{\text{Cu}(\text{L2})_2} > K_{\text{Zn}(\text{L2})_2} > K_{\text{Cd}(\text{L2})_2}$. From Table 1, the value of $K_{\text{M}(\text{L2})_2}/K_{\text{M}(\text{L1})_2}$ is in the order of Cu(II)>Cd(II)>Zn(II), which determines the coordination equilibrium. The greater the ratio is, the higher possibility of metal-ion selectivity will be, which is consistent with the experimental results of the heterometallic systems of Zn(II)/Cu(II) exhibited excellent metal-ion selectivity.

Table 1: The binding constants *K* of ligands **L1** and **L2** with metal ions Zn(II), Cd(II), and Cu(II).

<i>K</i>	L1	L2	L2/L1
Zn(II)	$3.8 \times 10^5 \text{ M}^{-1}$	$4.5 \times 10^5 \text{ M}^{-1}$	1.45
Cd(II)	$0.7 \times 10^5 \text{ M}^{-1}$	$2.4 \times 10^5 \text{ M}^{-1}$	3.43
Cu(II)	$2.4 \times 10^5 \text{ M}^{-1}$	$9.9 \times 10^5 \text{ M}^{-1}$	4.13

Except for the binding constants of model ligands with metal ions, single-crystal structures of **Cd₂L₂** and **Cu₂L₂** were obtained (Figure 3a–d and Tables S1 and S2). The diffraction-grade single crystals of **Cd₂L₂** and **Cu₂L₂** as white and dark green squares were obtained by slow vapor diffusion of isopropyl ether into an acetonitrile solution, which formed tight dimers with a head-to-tail coordination mode.^[81] The intramolecular distances between two layers of **Cd₂L₂** (3.26 Å) and **Cu₂L₂** (3.20 Å) were smaller than that of **Zn₂L₂** (3.51 Å)^[77] (Figure 3c and 3d). Furthermore, **Cu₂L₂** exhibited a relatively large twist degree of the pseudo-octahedral $\langle tpy - Cu(II) - tpy \rangle$ units (Figure 3b and 3d). Compared with **Zn₂L₂** and **Cd₂L₂**, the twist configuration further lead the increased length of Cu–N1 (2.315 Å), Cu–N2 (2.015 Å) and Cu–N3 (2.348 Å) coordination bonds, and compared with other complexes (Figure 3e and Table 2), the coordination bonds' lengths of **Cu₂L₂** were greatly changed (Figure S53). The smallest intramolecular distance and twist conformation of **Cu₂L₂** should be attributed to the multiple intramolecular π -stacking interactions (such as cation- π and π - π interactions) (Figures S51 and S52), which can provide extra force in the metallo-complexes based on **L2**. Combining the analysis of single-crystal structure analysis with binding constants, **Cu₂L₂** with

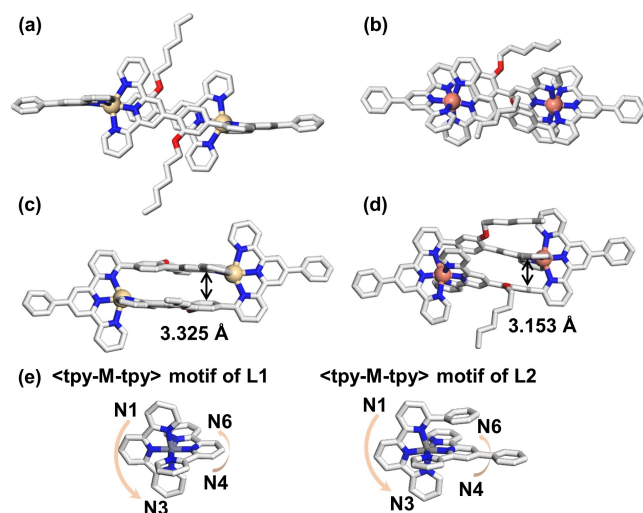


Figure 3. Single-crystal structures of **Cd₂L₂** (a and c), **Cu₂L₂** (b and d), and the sequences of coordinated nitrogen atoms (e).

Table 2: The coordination bond lengths of ligands **L1** and **L2** with metal ions Zn(II), Cd(II) and Cu(II).

d (Å)	d _{M–N1}	d _{M–N2}	d _{M–N3}	d _{M–N4}	d _{M–N5}	d _{M–N6}
ZnL1₂	2.193	2.069	2.192	2.178	2.091	2.217
CdL1₂	2.342	2.314	2.365	2.397	2.306	2.321
CuL1₂	2.194	1.963	2.194	2.194	1.963	2.194
Zn₂L₂	2.229	2.079	2.244	2.194	2.117	2.176
Cd₂L₂	2.337	2.296	2.379	2.373	2.320	2.356
Cu₂L₂	2.315	2.015	2.348	2.070	1.943	2.063

Note: **ZnL1₂**, **CdL1₂**, and **CuL1₂** are reported in Ref. [78–80]. **L1'** is 4'-(4-methoxyphenyl)-2,2':6',2''-terpyridine.

high coordination constant is the thermodynamic product, when co-assembly with **L1**, **L2**, Zn(II) and Cu(II).

DFT calculations. To further explore the plausible mechanism of the self-sorting and ultra-high metal-selectivity, the density functional theory (DFT) calculations were performed to investigate the metal-ligand interactions. The non-covalent interaction (NCI) method was utilized for visualizing and analyzing these interactions, revealing their types and providing quantitative insights into their strengths. Due to the existence of multiple intramolecular π -stacking interactions, two interaction models in metal complexes were proposed to figure out the key interactions for metal-ion selectivity (Figure S54). In model 1 (Figure S54a), the cation-N coordination interaction and cation- π interaction were considered. To eliminate the influence of π - π interactions, hypothetical mononuclear complexes of **L2** were used to calculate the cation- π interaction. The calculated binding energies (E_b) and NCI plots were presented in Figure 4. The calculated binding energies between metal cations and **L2** follow the order of **CuL2** > **ZnL2** > **CdL2**, consistent with the order of experimentally measured binding constants. The NCI plots revealed that there exist the cation-N coordination (similar in Metal-L1 interaction) and cation- π (metal interacting with the phenyl group positioned on one side of the pyridine) interactions. The cation- π interactions in the model system **CuL2** exhibits the relatively stronger attractive interaction than those in **ZnL2** and **CdL2** (as indicated in the red box), resulting in large binding energy in **CuL2**, thereby ligand **L2** inclinatively coordinated with Cu(II). In the model 2 (Figure S54b), cation-N coordination and π - π interactions coexist. The binding energies are in the order of **Zn₂L₂** > **Cd₂L₂** > **Cu₂L₂** (Figure S55), which is inconsistent with the order of binding constants. Thus, π - π interactions cannot be considered the determining factor for metal-ion selectivity. Consequently, the cation- π interactions were identified as playing a pivotal role in both the self-sorting behavior and metal ion selectivity.

In addition to the thermodynamic study, the kinetic analysis of metal-exchange process between **CuL1₂** and **Zn₂L₂** was also carried out to gain further insights into the reaction dynamics. The metal-exchange experiment was conducted at 50 °C for 9 hours and monitored using NMR spectroscopy. From the NMR spectra, the generation rate of **ZnL1₂** was determined by integrating the characteristic peaks corresponding to **ZnL1₂** to analyse metal-exchange kinetics. The metal-exchange reaction was completed at 6.25 h (Figure S14). The experimental data was found to be consistent with the fitting curve of a second-order reaction (Figure S15), which corresponded to a metal-exchange rate constant of 16.5 M⁻¹ min⁻¹. Comparison of binding constants of **ZnL1₂**, **CuL1₂**, **Cu₂L₂**, and **Zn₂L₂**, it was noted that the binding constant of **ZnL1₂** ($K = 3.8 \times 10^5$ M⁻¹) is 1.58-fold as that of **CuL1₂** ($K = 2.4 \times 10^5$ M⁻¹); the binding constant of **Cu₂L₂** ($K = 9.9 \times 10^5$ M⁻¹) is 2.2-fold as that of **Zn₂L₂** ($K = 4.5 \times 10^5$ M⁻¹). This result indicates the synergistic effect in metal-exchange reaction, suggesting that the model system has ultra-high metal-ion selectivity.

The self-assembly of ligand L3 with one type of metal ion. Based on the result of model ligands **L1** and **L2**, the

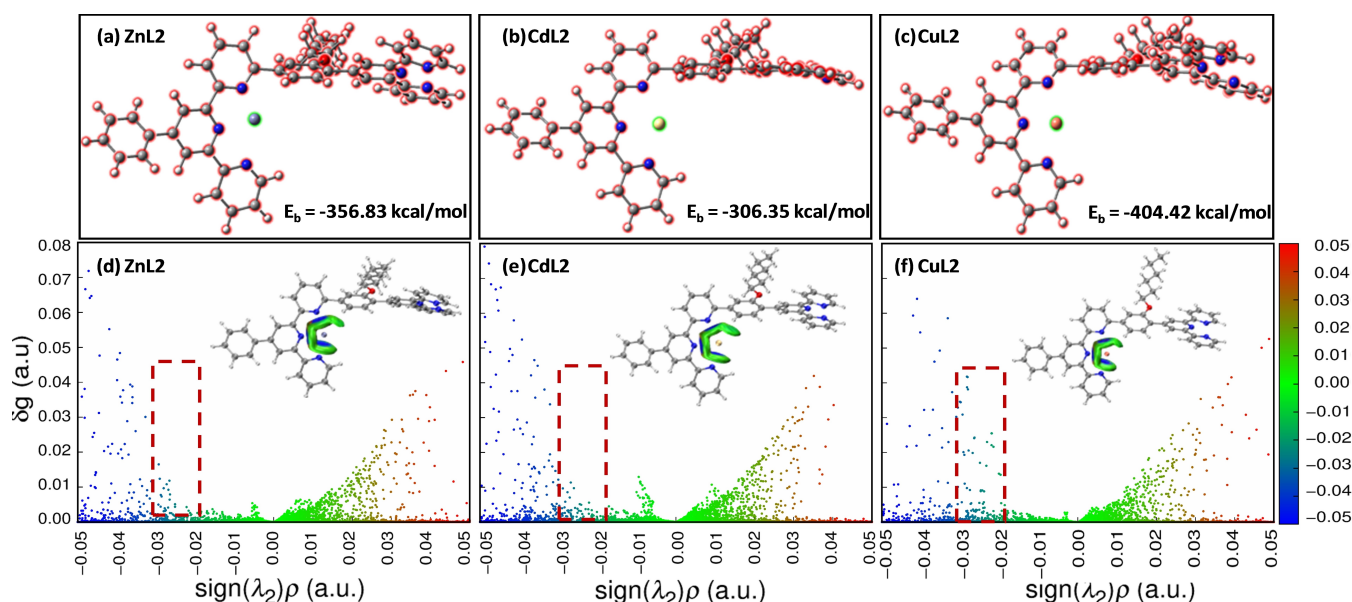


Figure 4. Binding energies of **ZnL2** (a), **CdL2** (b), and **CuL2** (c) and visualized NCI plots of **ZnL2** (d), **CdL2** (e), and **CuL2** (f) with a commonly used color scale to represent the attractive (blue) and repulsive (red) interactions.

tritopic terpyridyl ligand **L3** was designed by combining ligands **L1** and **L2** by Suzuki coupling reaction (Supporting Information Scheme S1, Figure S21–S37). The self-assembly of ligand **L3** with $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, $\text{Cd}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$, or $\text{Cu}(\text{NO}_3)_2 \cdot 2.5\text{H}_2\text{O}$ was carried out under an exact stoichiometric ratio of 2:3 in $\text{CHCl}_3/\text{MeOH}$ ($v/v=1:3$) at 50°C for 8 h, followed by addition of an excessive amount of NH_4PF_6 to obtain metallo-complexes.

Ligand **L3** and the metallo-complexes were initially characterized by multi-dimensional NMR spectroscopy (see details in Supporting Information). The ^1H NMR spectrum of **L3** displayed seven sets of proton signals for pyridine in Figure 5a, which were assigned with the aid of 2D NMR (Correlation Spectroscopy (COSY) and Nuclear Overhauser Effect Spectroscopy (NOESY), Figures S32–S34), due to the existence of three different terpyridyl motifs. Upon self-assembly, compared with ligand **L3**, the peaks of $\text{tpy-H}^{6,6'}$ in metallo-complexes with $\text{Zn}(\text{II})$ and $\text{Cd}(\text{II})$ showed a significant upfield shift due to the electronic shielding effect, indicating successful coordination of tpy units with $\text{Zn}(\text{II})$. Meanwhile, all proton signals of $\text{tpy-H}^{3,5'}$ exhibited diagnostic downfield shift (Figure 5a, 5b, Figures S38–S42). Interestingly, the signals corresponding to tpy-H of part b in metallocomplexes split into two peaks and became significantly broader. The broadening and splitting signals are likely due to the steric effects caused by the hexyl chains, which seriously restricts the rotation of substituted benzene rings, leading to the existence of different spatial directions of the hexyl chains.

ESI-MS and traveling-wave ion mobility mass spectrometry (TWIM-MS) were used to verify the chemical compositions of three complexes. ESI-MS showed only one prominent set of peaks with different charge states for **Zn₆L₃**, **Cd₆L₃**, and **Cu₆L₃**, resulting from the successive loss of different numbers of counterions (Figure 5c and Figure S45).

Upon deconvolution, the obtained chemical compositions of complexes agreed well with four **L3**, six metal ions, and twelve PF_6^- . Furthermore, the experimental isotope patterns matched well with the simulated values (Figures S46–48), confirming the exact composition of the complexes. TWIM-MS showed that each charge state of tetramers has a narrow drift-time distribution (Figure 5d), indicating the discrete and rigid assembly without other isomers or structural conformers.

To elucidate the self-assembly mechanism, as shown in Figure S6, ESI-MS spectra of the products self-assembled by ligand **L3** with $\text{Zn}(\text{II})$, $\text{Cd}(\text{II})$ and $\text{Cu}(\text{II})$ in a stoichiometric ratio of 4:4 showed three peaks corresponding to 2+ and 4+ charge states of **M₂L₃**, indicating that stable intermediates between ligand and metal ions were formed. With the continued addition of metal ions, the tetrameric helix **M₆L₃** was eventually obtained.

The self-assembly of dissymmetrical ligand L3 with two types of metal ions. Based on all above results, ligand **L3** was used to construct the heterometallic supramolecular structures. The self-assembly of ligand **L3** with $\text{Zn}(\text{II})$ and $\text{Cd}(\text{II})$ (an exact stoichiometric ratio of 2:1:2) was carried out under the same conditions (Figure 6a). ESI-MS was employed to characterize the obtained complex, as the proton signals of product were too complex and stacked in ^1H NMR spectrum. ESI-MS spectrum showed a series of peaks with different charge states from 4+ to 11+ (Figure 6b), the isotope patterns can be observed in each charge states. After deconvolution, the obtained molecular weights are a series of tetramers **Zn_nCd_{6-n}L₃** ($n=0-6$) containing different ratios of $\text{Zn}(\text{II})$ and $\text{Cd}(\text{II})$, which is similar to the assembly of **L3** and $\text{Cd}(\text{II})$ and $\text{Cu}(\text{II})$ (Figure S17).

Due to the self-sorting and metal-ion selectivity in the co-assembly with model ligands **L1** and **L2**, $\text{Zn}(\text{II})$ and $\text{Cu}(\text{II})$, **L3**, $\text{Zn}(\text{II})$ and $\text{Cu}(\text{II})$ (Figure 7a) are considered as

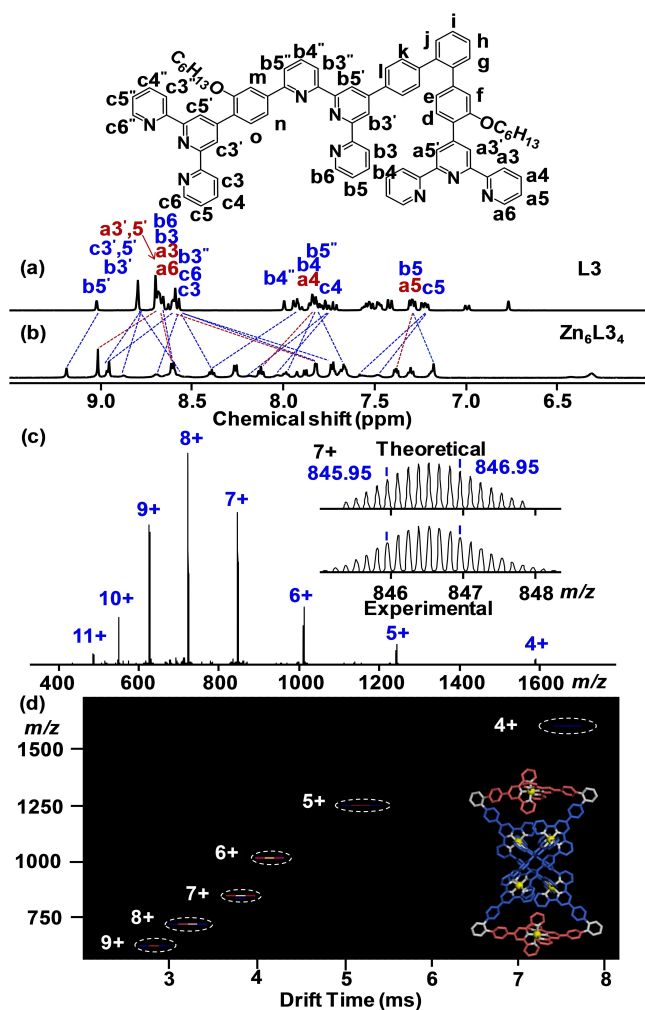


Figure 5. ¹H NMR spectra (500 MHz, 298 K) of (a) **L3** in CDCl₃ and (b) **Zn₆L₃₄** in CD₃CN; ESI-MS spectrum and TWIM-MS plots (*m/z* vs drift time) of **Zn₆L₃₄** (c, d).

the most promising candidate for the construction of the well-defined heterometallic supramolecule. ESI-MS spectrum showed only one prominent set of peaks with varying charge states resulting from the successive loss of different numbers of counterions, and only one isotope pattern was found in each charge state (Figure 7b and Figure S50). To avoid errors caused by extremely minor difference of atomic weights between Zn(II) (65.409 Da) and Cu(II) (63.546 Da), a stable internal standard metallo-complex **RuL₂** was used to correct the ESI-MS data. The *m/z* of **RuL₂** signal with 2+ charge state was close to that of **Zn₂Cu₄L₃₄** with 9+ charge state (Figures 7b and S18). As shown in Figure 6c, the corrected isotope pattern matched well with the simulated signal of four ligands **L3**, two Zn(II), and four Cu(II), indicating the successful construction of a well-defined bimetallic helical tetramer structure.

To further explore the location of metal ions, the assembly process of intermediates **Cu₂L₃₂** with metal and ligand ratio of 1:1 in stepwise method was studied via ¹H NMR and ESI-MS (Figures S19 and S20). From NMR spectra (Figures S19d and S20), the integral peak area

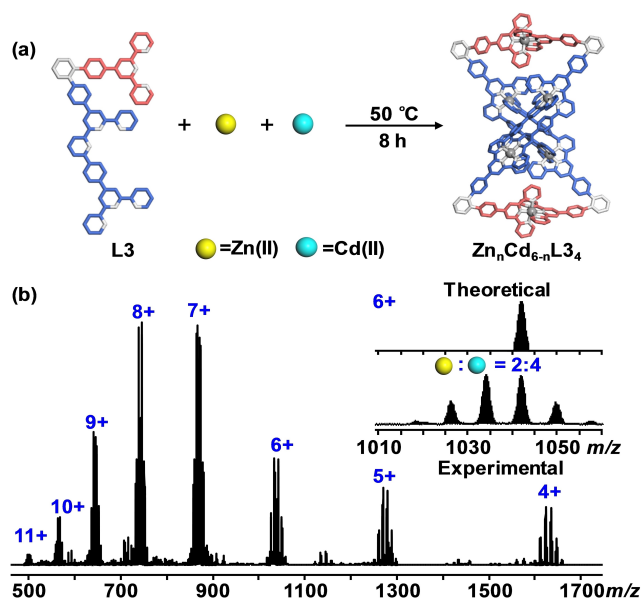


Figure 6. (a) Self-assembly of **L3**, Zn(II) and Cd(II) ions. (b) ESI-MS spectrum of **Zn₆Cd_{6-n}L₃₄**. The inset shows the experimental (bottom) and simulated (top) isotope pattern of **Zn₆Cd_{6-n}L₃₄** with the loss of 6 PF₆⁻ ions.

corresponded to the proton of uncoordinated terpyridyl motif of part A (red part), which indicated that the coordination parts were terpyridyl motifs of parts B and C (blue part). ESI-MS spectrum (Figure S19b) further confirmed there is only **Cu₂L₃₂** without other intermediates. Zn(II) was then added into the intermediates **Cu₂L₃₂** to obtain the heterometallic supramolecule **Zn₂Cu₄L₃₄**. The ¹H NMR and ESI-MS spectra were consistent with these of heterometallic supramolecule **Zn₂Cu₄L₃₄** in one-pot method (Figures 7b, S19c, and S19e). Though the proton signals of blue part of **Zn₂Cu₄L₃₄** are broadened due to the paramagnetic characterization of Cu(II), the proton signals of the red part coordinated with Zn(II) should be detectable. After analyzing ¹H and COSY NMR spectra (Figure S49), the peaks were assigned, corresponded well with the proton count of the red part, which provided valid evidence to confirm the metal sequence and site-selective self-assembly. However, all attempts to grow X-ray-quality single crystals have been proven to be unsuccessful, most likely due to existence of multiple alky chains. The optimized molecular model showed a helical tetrameric structure with Zn(II) and Cu(II) ions precisely anchored at the top/bottom and cross parts of the helicate, respectively (Figure 7a).

Conclusions

In summary, we proposed a novel intramolecular cation-π (ICπ) strategy to achieve ultra-high selectivity for different metal ions, and constructed the heterometallic metallo-supramolecule **Zn₂Cu₄L₃₄** encoding with precise sequence of metal ions via one-pot method in an error-free manner. Tritopic terpyridyl ligand **L3** was designed with model ligands **L1** and **L2**. After

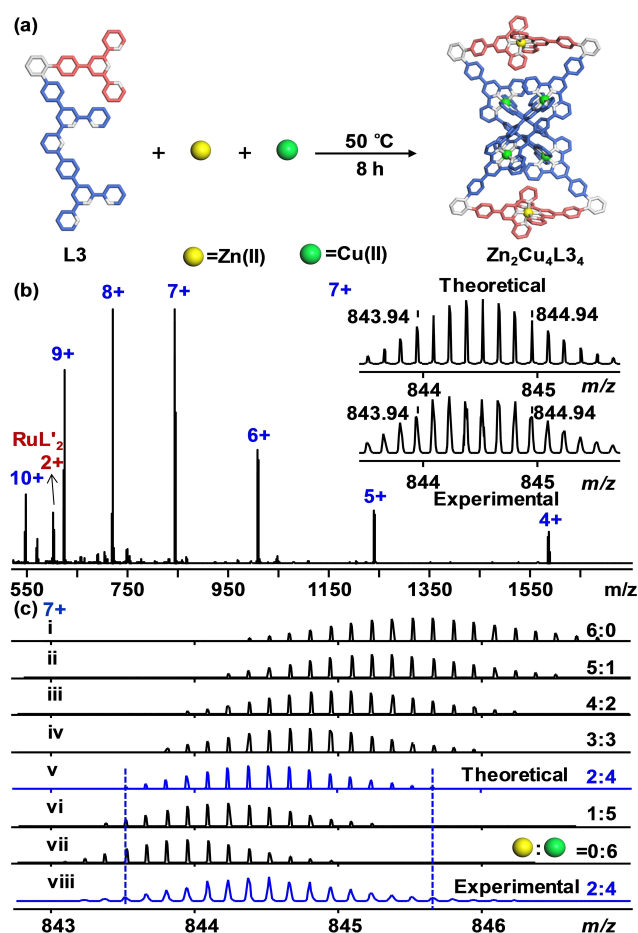


Figure 7. (a) Self-assembly of L3, Zn(II) and Cu(II). (b) ESI-MS spectrum of Zn₂Cu₄L₃₄. (c) Experimental isotope pattern (viii) and simulated isotope patterns (i-vii) of Zn(II), Cu(II) and ligand L3.

analyzing the self-assembly of L1 and L2 with Zn(II) and Cu(II) in one-pot reaction, only ZnL₁₂ and CuL₂₂ were formed, demonstrating the perfect self-sorting and ultra-high metal selectivity, induced by intramolecular cation- π interactions. This study proved a unique strategy to construct precise heterometallic supramolecules and even more complex structures only with one type motif. And it will also offer deeper insight into the potential functional materials.

Supporting Information

Synthetic details and characterizations of ligands and metallo-complexes including NMR, ESI-MS, TWIM-MS, and Single-crystal data (PDF). The authors have cited additional references within the Supporting Information.^[82-85]

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords: Heterometallic Architectures • Metal-Ion Selectivity • Self-Assembly • Supramolecular Chemistry • Terpyridine

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